Exp 6 : GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) AND OF PHASE MODULATED (PM) SIGNALS USING MATLAB

Objectives :

To generate and analyze Frequency Modulated (FM) and Phase Modulated (PM) signals using MATLAB by varying the instantaneous frequency and phase of a carrier signal in accordance with the message signal.

Code :

```
clc;
clear;
close all;

Am = 1;
Ac = 1;
fm = 5;
fc = 50;
beta = 5;
kp = 5;
fs = 1000;

t = 0:1/fs:1;

m = Am*cos(2*pi*fm*t); % Message signal
c = Ac*cos(2*pi*fc*t); % Carrier signal

fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
pm_sig = Ac*cos(2*pi*fc*t + kp*m); % PM

Df = beta*fm; % Frequency deviation

fprintf('Frequency Deviation = %d Hz\n',Df);
```

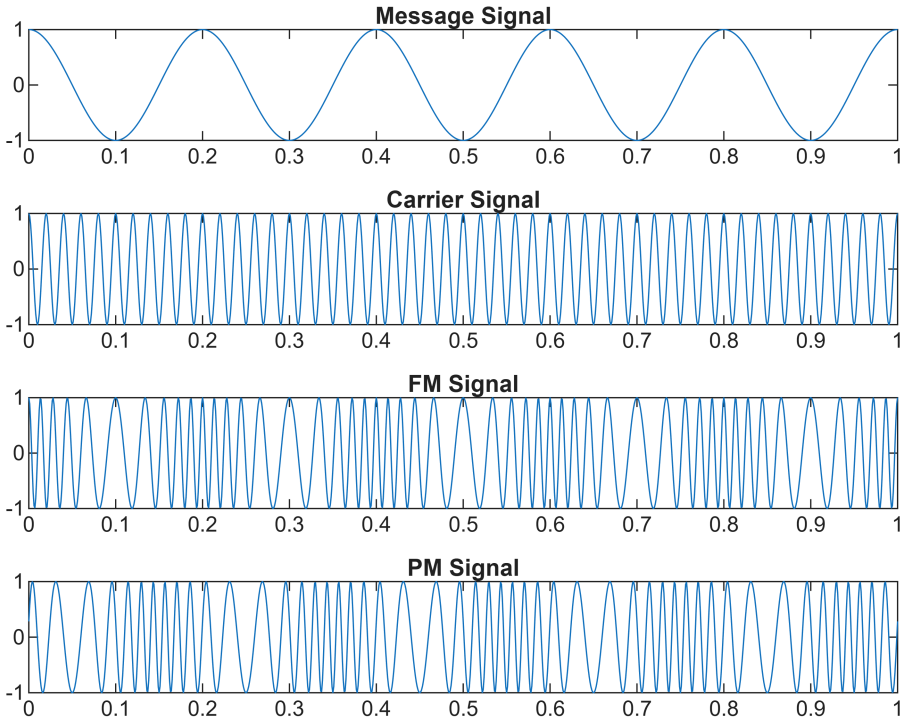
```
Frequency Deviation = 25 Hz
```

```
fprintf('FM Modulation Index = %d\n',beta);
```

```
FM Modulation Index = 5
```

```


subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
subplot(4,1,4), plot(t,pm_sig), title('PM Signal')
```



Result :

The Frequency Modulated (FM) and Phase Modulated (PM) signals were successfully generated using MATLAB by varying the instantaneous frequency and phase of the carrier signal according to the message signal.

• Frequency Deviation (Δf) = 25 Hz

• FM Modulation Index (β) = 5